```python
from googleapiclient.discovery import build
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker
import matplotlib.ticker as mticker
import matplotlib.font_manager as fm
import matplotlib as mpl
from matplotlib import rcParams
import pandoc

api_key = 'AIzaSyB6z5v2sgUzCYD6N-CRTON-302-Z_3nzQY'
channel_ids = ['UCKE_BzSxej1BM67qt8PW6tw',
               'UCeQ7hsZV1PxX51s78Iciz5A',
               'UCvaVvyzE7gC0sI-3PZ_vYFQ',
               'UCcFb_EZCGhxPOgzsK2eUe6w'
              ]

youtube = build('youtube', 'v3', developerKey=api_key)
```

Function to get Channel Statistics

```python
def get_channel_stats(youtube, channel_ids):
    all_data = []
    request = youtube.channels().list(
              part='snippet,contentDetails,statistics',
id=channel_ids)
    response = request.execute()

    for i in range(len(response['items'])):
        data = dict(channel_name = response['items'][i]['snippet']
['title'],
                    subscribers = response['items'][i]['statistics']
['subscriberCount'],
                    view_counts = response['items'][i]['statistics']
['viewCount'],
                    total_videos = response['items'][i]['statistics']
['videoCount'],
                    playlist_id = response['items'][i]['contentDetails']
['relatedPlaylists']['uploads'])
        all_data.append(data)

    return all_data

channel_stats = get_channel_stats(youtube, channel_ids)

channel_data = pd.DataFrame(channel_stats)
channel_data
```

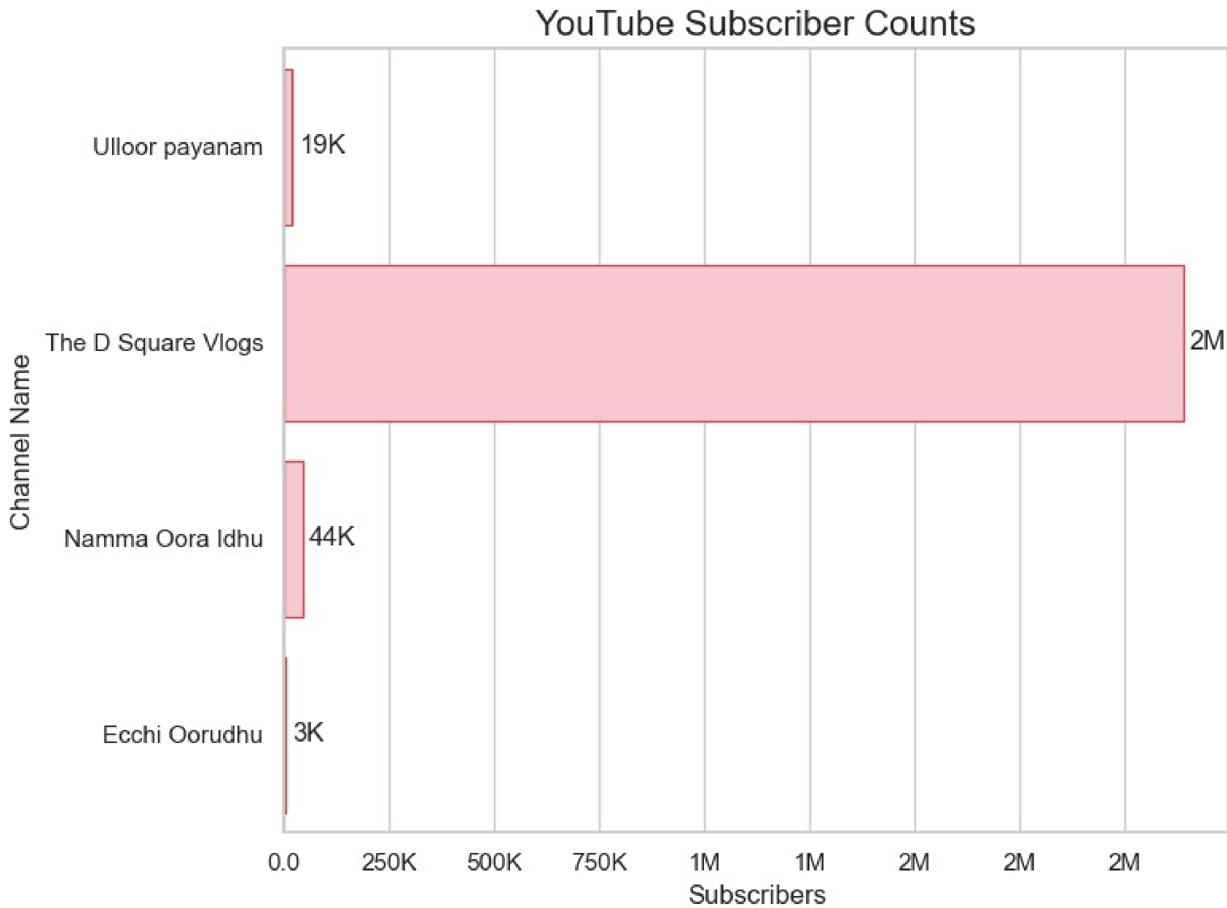
```
         channel_name subscribers view_counts total_videos  \
0      Ulloor payanam       18600    15078361          128
1  The D Square Vlogs     2140000  1649715804         1175
2     Namma Oora Idhu       43600     4828588          241
3       Ecchi Oorudhu        3440      268395           64

                playlist_id
0  UUvaVvyzE7gC0sI-3PZ_vYFQ
1  UUeQ7hsZV1PxX51s78Iciz5A
2  UUcFb_EZCGhxPOgzsK2eUe6w
3  UUKE_BzSxej1BM67qt8PW6tw
```

```
channel_data["subscribers"] =
channel_data["subscribers"].astype("int")
channel_data["view_counts"] =
channel_data["view_counts"].astype("int")
channel_data["total_videos"] =
channel_data["total_videos"].astype("int")
```

Youtube Subscriber Counts

```
def format_large_number(val):
    if val >= 1_000_000_000:
        return f'{val / 1_000_000_000:.1f}B'
    elif val >= 1_000_000:
        return f'{val / 1_000_000:.0f}M'
    elif val >= 1_000:
        return f'{val / 1_000:.0f}K'
    else:
        return str(val)

plt.figure(figsize=(8, 6))
sns.set(style='whitegrid')
bar = sns.barplot(x='subscribers', y='channel_name',
data=channel_data, color='pink', edgecolor = 'r')

bar.xaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _:
format_large_number(x)))

for container in bar.containers:
        labels = [format_large_number(bar.get_width()) for bar in
container]
        bar.bar_label(container, labels=labels, label_type='edge',
padding=3)

plt.title('YouTube Subscriber Counts', fontsize=16)
plt.ylabel('Channel Name')
```

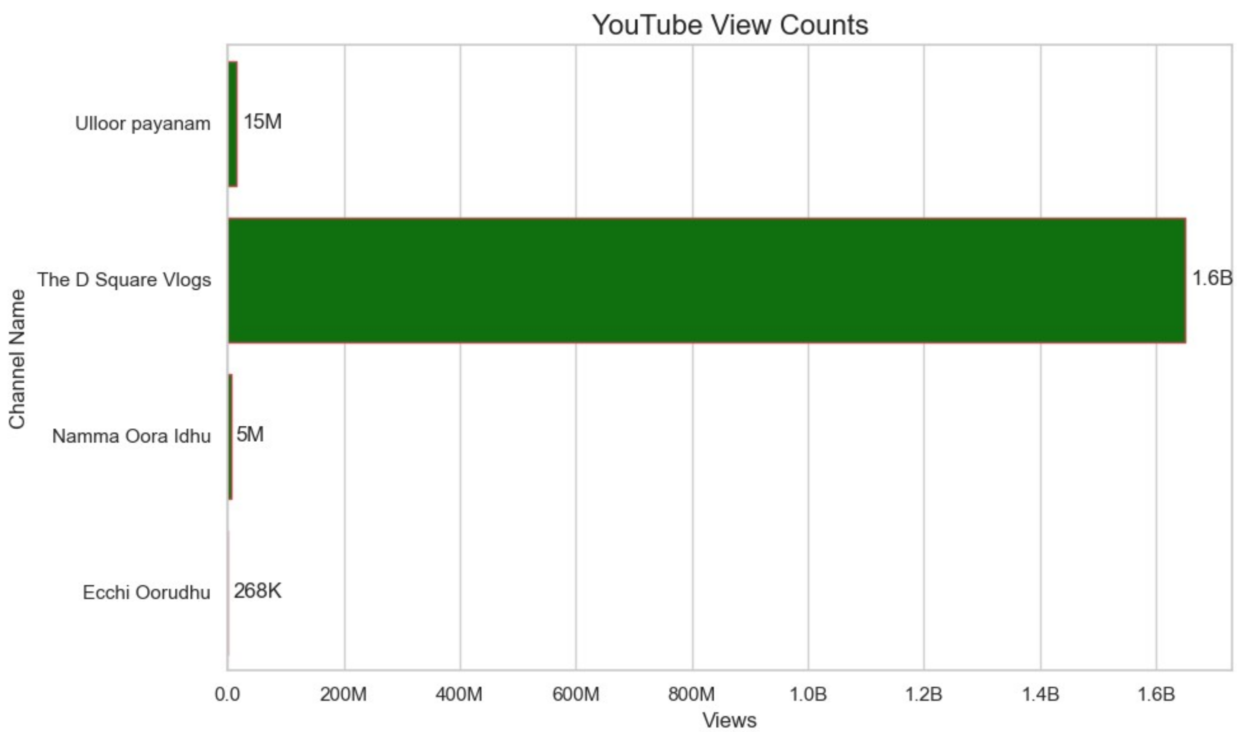
```
plt.xlabel('Subscribers')
plt.tight_layout()
plt.show()
```



Youtube View Counts

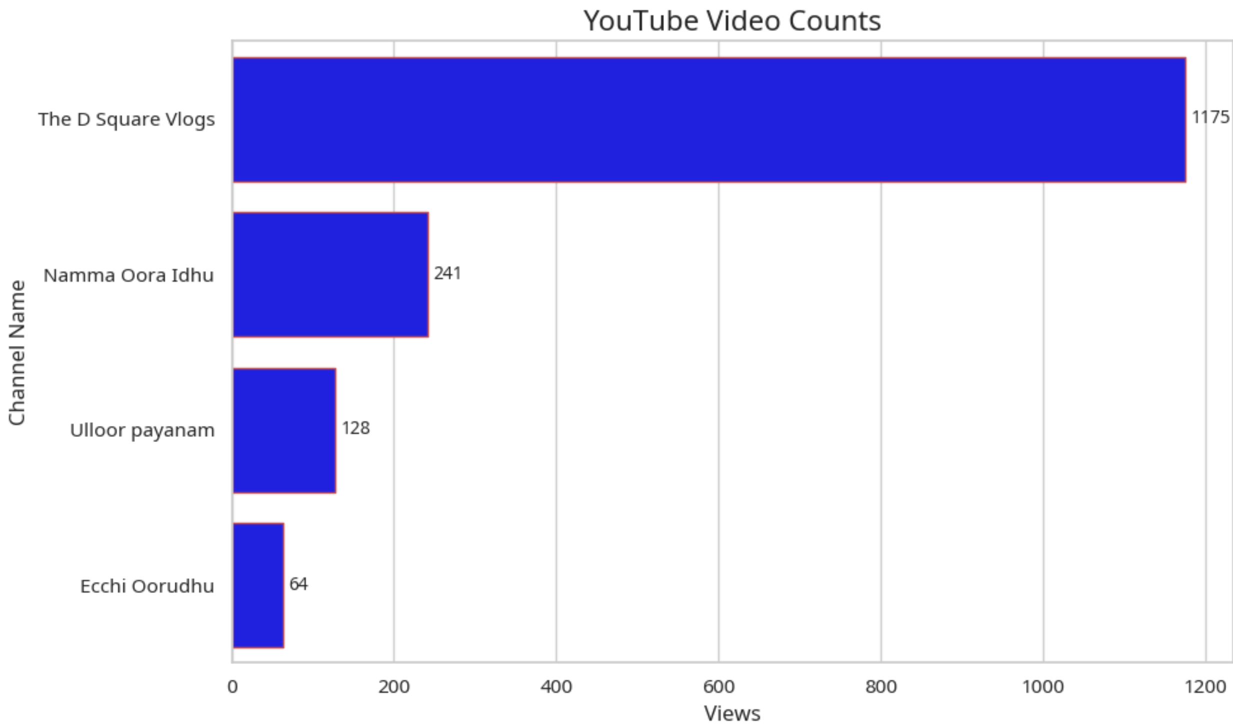
```
def format_large_number(val):
    if val >= 1_000_000_000:
        return f'{val / 1_000_000_000:.1f}B'
    elif val >= 1_000_000:
        return f'{val / 1_000_000:.0f}M'
    elif val >= 1_000:
        return f'{val / 1_000:.0f}K'
    else:
        return str(val)

plt.figure(figsize=(10, 6))
sns.set(style='whitegrid')
bar = sns.barplot(x='view_counts', y='channel_name',
data=channel_data, color='green', edgecolor = 'r')
```

```
bar.xaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _:
format_large_number(x)))

for container in bar.containers:
        labels = [format_large_number(bar.get_width()) for bar in
container]
        bar.bar_label(container, labels=labels, label_type='edge',
padding=3)

plt.title('YouTube View Counts', fontsize=16)
plt.ylabel('Channel Name')
plt.xlabel('Views')
plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(10, 6))
#sns.set(style='whitegrid')
channel_data_sorted = channel_data.sort_values('total_videos',
ascending = False)
#channel_data = channel_data.sort_values(by='channel_name',
ascending=True)
#plt.bar(channel_data_sorted['total_videos'],
channel_data_sorted['channel_name'], color='blue', edgecolor = 'r')

bar1 = sns.barplot(y='channel_name', x='total_videos',
data=channel_data_sorted, color='blue', edgecolor = 'r')
```

```
for container in bar1.containers:
    bar1.bar_label(container, padding=3, fontsize=10)

plt.title('YouTube Video Counts', fontsize=16)
plt.ylabel('Channel Name')
plt.xlabel('Views')
plt.tight_layout()
plt.show()
plt.show()
```




```
playlist_id = channel_data.loc[channel_data['channel_name'] == 'Ecchi Oorudhu', 'playlist_id'].iloc[0]
```

Function to get Video ID

```
def get_video_ids(youtube, playlist_id):
    request = youtube.playlistItems().list(
              part='contentDetails', playlistId = playlist_id,
              maxResults = 50)
    response = request.execute()

    video_ids = []
```

```
    for i in range(len(response['items'])):
        video_ids.append(response['items'][i]['contentDetails']
['videoId'])

    next_page_token = response.get('nextPageToken')
    more_pages = True

    while more_pages:
        if next_page_token is None:
            more_pages = False
        else:
            request = youtube.playlistItems().list(
                      part='contentDetails', playlistId = playlist_id,
                      maxResults = 50,
                      pageToken = next_page_token)
            response = request.execute()

            for i in range(len(response['items'])):
                video_ids.append(response['items'][i]
['contentDetails']['videoId'])

            next_page_token = response.get('nextPageToken')

    return video_ids

video_ids = get_video_ids(youtube, playlist_id)
```

Function to get video details

```
def get_video_details(youtube, video_ids):
    all_video_stats = []
    for i in range(0, len(video_ids), 50):
        request = youtube.videos().list(
                    part='snippet, statistics', id =
','.join(video_ids[i:i+50]))
        response = request.execute()

        for video in response['items']:
            video_stats = dict(Title = video['snippet']['title'],
                               Published_date = video['snippet']
['publishedAt'],
                               Views = video['statistics']
['viewCount'],
                               Likes = video['statistics']
['likeCount'],
                               Comments = video['statistics']
['commentCount'])
            all_video_stats.append(video_stats)
    return all_video_stats
```

```
video_details = get_video_details(youtube, video_ids)

video_data = pd.DataFrame(video_details)
video_data

                                                Title
Published_date  \
0   We are Back !!! @EcchiOorudhu 2022 | Arambikal...  2022-06-
28T11:35:59Z
1   WINSOR CASTLE SALEM | SWIMMING POOL SALEM | MU...  2021-09-
13T14:22:43Z
2   FAMOUS PANIPOORI SHOP SALEM😋 | பானிபூரி கடை சே...  2021-08-
31T14:10:43Z
3   YERCAUD TANDOORI TEA | YERCAUD VLOG | YERCAUD ...  2021-08-
28T12:30:11Z
4   For IDLY LOVERS😋 |VARIETIES OF ILDY | IDLY MAN...  2021-08-
03T08:35:29Z
..                                                ...
...
59  Dum Parotta  Salem | தம் பரோட்டா சேலம் | Highw...  2021-01-
09T14:34:01Z
60  Kaadai muttai Salem | காடை முட்டை | Quail Egg ...  2021-01-
04T07:47:22Z
61  Dosai Kadai in Salem | தோசை கடை | Variety of D...  2020-12-
30T09:39:29Z
62  Ice Cream Spot in Salem | Most Satisfying Ice ...  2020-12-
24T07:46:30Z
63  Kal Meen Fry in Salem | கல் மீன் ஃப்ரை சேலம் |...  2020-12-
19T09:30:56Z

    Views Likes Comments
0     537    37        7
1    1726    74        4
2    2223   117        5
3     888    40       13
4    1529    75        5
..    ...   ...      ...
59   3904   198       16
60   3448   173       20
61  18681   392       33
62   3256   208       22
63   7520   342       37

[64 rows x 5 columns]

video_data["Views"] = video_data["Views"].astype("int")
video_data["Likes"] = video_data["Likes"].astype("int")
video_data["Comments"] = video_data["Comments"].astype("int")
video_data["Published_date"] =
pd.to_datetime(video_data["Published_date"]).dt.date
```

```
video_data
```

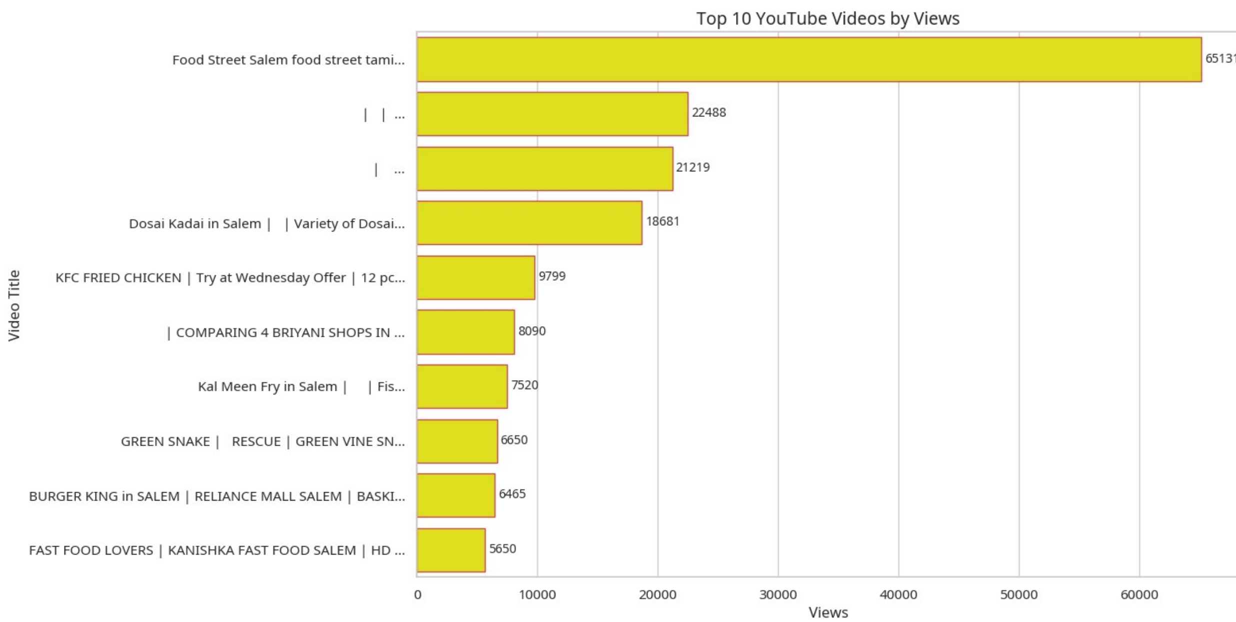
```
                                                Title Published_date
Views  \
0   We are Back !!! @EcchiOorudhu 2022 | Arambikal...     2022-06-28
537
1   WINSOR CASTLE SALEM | SWIMMING POOL SALEM | MU...     2021-09-13
1726
2   FAMOUS PANIPOORI SHOP SALEM😋 | பானிபூரி கடை சே...     2021-08-
31   2223
3   YERCAUD TANDOORI TEA | YERCAUD VLOG | YERCAUD ...     2021-08-28
888
4   For IDLY LOVERS😋 |VARIETIES OF ILDY | IDLY MAN...     2021-08-03
1529
..                                                ...            ...
...
59  Dum Parotta  Salem | தம் பரோட்டா சேலம் | Highw...     2021-01-09
3904
60  Kaadai muttai Salem | காடை முட்டை | Quail Egg ...     2021-01-04
3448
61  Dosai Kadai in Salem | தோசை கடை | Variety of D...     2020-12-
30  18681
62  Ice Cream Spot in Salem | Most Satisfying Ice ...     2020-12-24
3256
63  Kal Meen Fry in Salem | கல் மீன் ஃப்ரை சேலம் |...     2020-12-19
7520

    Likes  Comments
0      37         7
1      74         4
2     117         5
3      40        13
4      75         5
..    ...       ...
59    198        16
60    173        20
61    392        33
62    208        22
63    342        37

[64 rows x 5 columns]
```

```
top_10_videos = video_data.sort_values(by='Views', ascending = False).head(10)

top_10_videos
```

```
                                                Title Published_date
Views  \
50  சேலத்தின் மினி Food Street Salem food street ...     2021-03-04
```

```
65131
40  இட்லி கறி கொத்து | முட்டை லாபா | வித்தியாசமான ...     2021-
04-15  22488
43  தேவி அக்கா கடை சேலம் | விலை குறைவில் திருப்திய...     2021-
04-03  21219
61  Dosai Kadai in Salem | தோசை கடை | Variety of D...     2020-12-
30  18681
14  KFC FRIED CHICKEN | Try at Wednesday Offer | 1...     2021-07-04
9799
28  சேலம் பிரியாணி கடை | COMPARING 4 BRIYANI SHOPS...     2021-05-
25   8090
63  Kal Meen Fry in Salem | கல் மீன் ஃப்ரை சேலம் |...     2020-12-19
7520
22  GREEN SNAKE | பச்சை பாம்பு RESCUE | GREEN VIN...     2021-06-11
6650
6   BURGER KING in SALEM | RELIANCE MALL SALEM | B...     2021-07-28
6465
17  FAST FOOD LOVERS | KANISHKA FAST FOOD SALEM |...     2021-06-25
5650

    Likes  Comments
50    959        80
40    643        57
43    355        30
61    392        33
14    174        10
28    194        32
63    342        37
22     86        15
6     193        18
17    259        11

font_path = 'NotoSansTamil-Regular.ttf'

tamil_font = fm.FontProperties(fname=font_path)
mpl.rcParams['font.family'] = tamil_font.get_name()


top_videos = top_10_videos.sort_values(by='Views', ascending=True)

top_videos['short_title'] = top_10_videos['Title'].apply(lambda x:
x[:50] + '...' if len(x) > 50 else x)

top_videos['short_title'] =
top_videos['short_title'].str.replace(r'[^\x00-\x7F]+', '',
regex=True)

plt.figure(figsize=(12, 8))

ax1 = sns.barplot(data = top_videos, x = 'Views', y = 'short_title',
```

```
color = 'yellow', edgecolor = 'r')

#bars = plt.bar(top_videos['Views'], top_videos['short_title'],
color='yellow', edgecolor = 'r')
plt.gca().invert_yaxis()

for container in ax1.containers:
    ax1.bar_label(container, padding=3, fontsize=10)

plt.title("Top 10 YouTube Videos by Views", fontsize=14)
plt.xlabel("Views")
plt.ylabel("Video Title")
plt.show()
```




```
top_10_liked_videos = video_data.sort_values(by='Likes', ascending =
False).head(10)

top_10_liked_videos

                                                Title Published_date
Views  \
50  சேலத்தின் மினி Food Street Salem food street ...     2021-03-04
65131
40  இட்லி கறி கொத்து | முட்டை லாபா | வித்தியாசமான ...     2021-
04-15  22488
61  Dosai Kadai in Salem | தோசை கடை | Variety of D...     2020-12-
30  18681
43  தேவி அக்கா கடை சேலம் | விலை குறைவில் திருப்திய...     2021-
04-03  21219
63  Kal Meen Fry in Salem | கல் மீன் ஃப்ரை சேலம் |...     2020-12-19
```

```
7520
17  FAST FOOD LOVERS | KANISHKA FAST FOOD SALEM |...     2021-06-25
5650
62  Ice Cream Spot in Salem | Most Satisfying Ice ...     2020-12-24
3256
59  Dum Parotta  Salem | தம் பரோட்டா சேலம் | Highw...     2021-01-09
3904
28  சேலம் பிரியாணி கடை | COMPARING 4 BRIYANI SHOPS...     2021-05-
25   8090
6   BURGER KING in SALEM | RELIANCE MALL SALEM | B...     2021-07-28
6465

    Likes  Comments
50    959        80
40    643        57
61    392        33
43    355        30
63    342        37
17    259        11
62    208        22
59    198        16
28    194        32
6     193        18
```

```
font_path = 'NotoSansTamil-Regular.ttf'

tamil_font = fm.FontProperties(fname=font_path)
mpl.rcParams['font.family'] = tamil_font.get_name()


top_videos = top_10_videos.sort_values(by='Likes', ascending=True)

top_videos['short_title'] = top_10_videos['Title'].apply(lambda x:
x[:50] + '...' if len(x) > 50 else x)

top_videos['short_title'] =
top_videos['short_title'].str.replace(r'[^\x00-\x7F]+', '',
regex=True)

plt.figure(figsize=(12, 8))

ax1 = sns.barplot(data = top_videos, x = 'Likes', y = 'short_title',
color = 'green', edgecolor = 'r')

#bars = plt.bar(top_videos['Views'], top_videos['short_title'],
color='yellow', edgecolor = 'r')
plt.gca().invert_yaxis()

for container in ax1.containers:
    ax1.bar_label(container, padding=3, fontsize=10)
```

```
plt.title("Top 10 YouTube Videos by Likes", fontsize=14)
plt.xlabel("Likes")
plt.ylabel("Video Title")
plt.show()
```